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Author(s): JAMES G. MACKINNON and MATTHEW D. WEBB

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WILD BOOTSTRAP INFERENCE FOR WILDLY DIFFERENT CLUSTER SIZES

JAMES G. MACKINNON^{a*} AND MATTHEW D. WEBB^b

^a *Department of Economics, Queen's University, Kingston, Ontario, Canada*

^b *Department of Economics, Carleton University, Ottawa, Ontario, Canada*

SUMMARY

The cluster robust variance estimator (CRVE) relies on the number of clusters being sufficiently large. Monte Carlo evidence suggests that the ‘rule of 42’ is not true for unbalanced clusters. Rejection frequencies are higher for datasets with 50 clusters proportional to US state populations than with 50 balanced clusters. Using critical values based on the wild cluster bootstrap performs much better. However, this procedure fails when a small number of clusters is treated. We explain why CRVE *t* statistics and the wild bootstrap fail in this case, study the ‘effective number’ of clusters and simulate placebo laws with dummy variable regressors. Copyright © 2016 John Wiley & Sons, Ltd.

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Supporting information may be found in the online version of this article.

1. INTRODUCTION

Many empirical papers use data that are clustered or grouped. This clustering causes problems for inference whenever there is intra-cluster correlation, especially when there are independent variables that are constant within groups. This problem has been known since Kloeck (1981) and Moulton (1990), and many procedures have been developed to deal with the tendency for intra-cluster correlation to bias standard errors downwards. The most common procedure is the cluster robust variance estimator (CRVE), which uses a formula (see Section 2) proposed in several papers, of which the earliest may be Liang and Zeger (1986). Stata uses this estimator when the `cluster` command is invoked.

The CRVE has been shown to work well when the number of clusters is large. However, several papers have pointed out problems with the estimator when the number of clusters is small. General results on covariance matrix estimation in White (1984) imply that the CRVE is consistent under three key assumptions:

- A1. The number of clusters goes to infinity.
- A2. The within-cluster error correlations are the same for all clusters.
- A3. Each cluster contains an equal number of observations.

The limitations of the CRVE when assumption A1 is poor are now well known (see, among others, Bertrand *et al.*, 2004; Donald and Lang, 2007; Brewer *et al.*, 2013). A wild bootstrap procedure that often works well when the number of clusters is not too small was proposed by Cameron *et al.* (2008). It was modified to handle cases with 12 or fewer clusters by Webb (2014). Assumptions A2 and A3 were relaxed by Carter *et al.* (2017), which also showed how to calculate the ‘effective number’

* Correspondence to: James G. MacKinnon, Department of Economics, Queen's University, Kingston, Ontario, Canada E-mail: jgm@econ.queensu.ca

of clusters for cases with heterogeneous within-cluster correlation and unequal (unbalanced) cluster sizes. Cameron and Miller (2015) provide a thorough recent survey of cluster robust inference.

Assumption A3 is particularly important. Previous Monte Carlo experiments on the effectiveness of the CRVE, notably those in Bertrand *et al.* (2004) and Cameron *et al.* (2008), primarily use datasets with equal-sized clusters. Both papers also perform experiments with data from the Current Population Survey (CPS), as discussed in Section 7. Most of the simulations in the former paper use data averaged over state–year pairs. This imposes assumption A3, because each state has only one observation per year. Some simulations do involve micro data (unbalanced, clustered by state or state–year pair), but rejection rates with clustering at the state level are not reported.

Previous results have led to the rule of thumb that the CRVE works reasonably well when the number of clusters is sufficiently large. Angrist and Pischke (2008) suggest that 42 clusters are enough for reliable inference. However, we show that the ‘rule of 42’ no longer holds when the assumption of equal-sized clusters is relaxed. Inference using CRVE standard errors can be quite unreliable even with 100 unbalanced clusters.

Many real-world datasets have wildly unequal cluster sizes. American datasets clustered at the state level are a prime example. A dataset with observations in clusters proportional to current state populations will have 12% of the sample from California. Eleven states will each contain less than 0.5% of the total sample, and the largest cluster will be roughly 60 times the size of the smallest one. This is a severe violation of the assumption of equal-sized clusters.

The remainder of the paper is organized as follows. The next two sections briefly discuss the two methods that we investigate, which promise improved inference with clustered data. Section 2 describes the wild cluster bootstrap and Section 3 discusses the use of critical values for CRVE t -statistics based on the effective number of clusters.

Section 4 presents Monte Carlo evidence using simulated datasets with a continuous test regressor and either equal cluster sizes or ones proportional to state populations. We show that inference based on CRVE t -statistics can perform poorly in the latter case. Using critical values based on the effective number of clusters instead of the actual number usually improves matters, but it does not always yield reliable inferences. In contrast, the wild cluster bootstrap procedure always performs extremely well.

The remainder of the paper deals with estimating treatment effects, mainly in the context of difference-in-differences (DiD) estimates. For treatment effects, cluster sizes matter, but the number of treated clusters matters even more. The wild bootstrap works very well in most cases, but all the methods fail badly when the number of treated clusters is small or (in some cases) large. Section 5 presents Monte Carlo evidence for the DiD case using simulated datasets. Additional simulation results for the pure treatment case are presented in Section A.3 of the online Appendix (supporting information).

Section 6 explains why inference based on CRVE t -statistics, with or without the wild bootstrap, can fail with few treated clusters. The theoretical results of this section accord remarkably well with our simulation results. Section 7 extends the ‘placebo laws’ Monte Carlo experiments of Bertrand *et al.* (2004). Section 8 contains a brief empirical example based on Angrist and Kugler (2008) and Section 9 concludes.

2. THE WILD CLUSTER BOOTSTRAP

A linear regression model with clustered errors may be written as

$$y \equiv \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_G \end{bmatrix} = X\beta + \epsilon \equiv \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_G \end{bmatrix} \beta + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_G \end{bmatrix} \quad (1)$$

where each cluster, indexed by g , has N_g observations. The matrix $\mathbf{X}$ and the vectors $\mathbf{y}$ and $\boldsymbol{\epsilon}$ have $N = \sum_{g=1}^G N_g$ rows, $\mathbf{X}$ has k columns, and the parameter vector $\boldsymbol{\beta}$ has k rows. The covariance matrix of $\boldsymbol{\epsilon}$ is $\boldsymbol{\Omega}$, an $N \times N$ block-diagonal matrix with G diagonal blocks $\boldsymbol{\Omega}_g$, each of them $N_g \times N_g$, corresponding to the G clusters.¹ Ordinary least squares (OLS) estimation of equation (1) yields estimates $\hat{\boldsymbol{\beta}}$ and residuals $\hat{\boldsymbol{\epsilon}}$. There are several CRVEs. The most popular CRVE, which we investigate, appears to be

$$\frac{G(N-1)}{(G-1)(N-k)} (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}'_g \hat{\boldsymbol{\epsilon}}_g \hat{\boldsymbol{\epsilon}}'_g \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1} \quad (2)$$

The first factor here is asymptotically negligible, but it always makes the CRVE larger when G and N are finite.

We wish to test the hypothesis that a single coefficient is zero. Without loss of generality, we let this be the last one, β_k . The procedure for using the (restricted) wild cluster bootstrap of Cameron *et al.* (2008) to test the hypothesis that $\beta_k = 0$ is

1. Estimate equation (1) by OLS.
2. Calculate t_k , the t -statistic for $\beta_k = 0$, using the square root of the k th diagonal element of expression (2) as a cluster robust standard error.
3. Re-estimate the model (1) subject to the restriction that $\beta_k = 0$, so as to obtain the restricted residuals $\tilde{\boldsymbol{\epsilon}}$ and the restricted estimates $\tilde{\boldsymbol{\beta}}$.
4. For each of B bootstrap replications, indexed by j , generate a new set of bootstrap dependent variables y_{ig}^{*j} using the bootstrap DGP:

$$y_{ig}^{*j} = X_{ig} \tilde{\boldsymbol{\beta}} + \tilde{\epsilon}_{ig} v_g^{*j} \quad (3)$$

where y_{ig}^{*j} is an element of the vector $\mathbf{y}^{*j}$ of observations on the bootstrap dependent variable, X_{ig} is the corresponding row of $\mathbf{X}$, and so on. Here v_g^{*j} is a random variable that follows the Rademacher distribution (see Davidson and Flachaire, 2008). It takes the values 1 and -1 with equal probability. Note that we would not want to use the Rademacher distribution if G were smaller than about 12; see Webb (2014), which proposes an alternative for such cases.

5. For each bootstrap replication, estimate equation (1) using $\mathbf{y}^{*j}$ as the regressand, and calculate t_k^{*j} , the bootstrap t statistic for $\beta_k = 0$, using the square root of the k th diagonal element of equation (2), with bootstrap residuals replacing OLS ones, as the standard error.
6. Calculate the symmetric bootstrap P -value:

$$\hat{p}_s^* = \frac{1}{B} \sum_{j=1}^B I(|t_k^{*j}| > |t_k|) \quad (4)$$

It would also be valid to use an equal-tail P -value. In our simulations, symmetric and equal-tail P -values were always extremely close.

The wild cluster bootstrap procedure described here has two key features. The first is that the bootstrap error terms for every observation in cluster g depend on the same random variable v_g^{*j} . This

¹ The (excessively strong) conditions of White (1984) imply that the matrices $\mathbf{X}'_g \boldsymbol{\Omega}_g \mathbf{X}_g$ must be identical across clusters, which is unlikely to hold in practice.

ensures that, to the extent that the residuals $\tilde{\epsilon}_{ig}$ preserve the variances and within-cluster covariances of the error terms ϵ_{ig} , the bootstrap DGP also preserves these properties. However, this feature can cause serious problems in certain cases; see Section 6. The second key feature is that the bootstrap DGP (3) uses estimates under the null hypothesis. It would also be valid to use the unrestricted residuals $\hat{\epsilon}_{ig}$ instead of the restricted ones $\tilde{\epsilon}_{ig}$ in equation (3), and we discuss that variant in Section 6. The two variants of the wild cluster bootstrap can yield dramatically different results when the number of treated clusters is small.

3. THE EFFECTIVE NUMBER OF CLUSTERS

The most obvious way to perform a test using a CRVE t statistic based on (2) is to compare it with the Student's t distribution with $N - k$ degrees of freedom. However, it is well known that this procedure almost always over-rejects. It is generally much better to use the $t(G - 1)$ distribution, as suggested by Donald and Lang (2007) and Bester *et al.* (2011). However, it may be possible to do even better if the degrees-of-freedom parameter is chosen in a more sophisticated way.

Carter *et al.* (2017), hereafter referred to as CSS, proposes a method for estimating the 'effective number' of clusters, G^* . This number depends in a fairly complicated way on the X_g matrices, the cluster sizes N_g , $g = 1, \dots, G$, and a parameter ρ that measures within-cluster correlation. CSS focuses on the use of G^* as a diagnostic, suggesting that inference may be unreliable when G and G^* differ substantially. However, the paper also mentions, but does not investigate, the possibility of using critical values from the $t(G^*)$ distribution together with conventional CRVE t statistics for inference. By analogy with the recommendation of Donald and Lang (2007), it seems more natural to use the $t(G^* - 1)$ distribution. We investigate both procedures.

CSS suggest setting $\rho = 1$. Alternatively, one can estimate ρ , which can be done in several ways. One of them is to use a method that is standard in the literature on panel data (see Davidson and MacKinnon, 2004, Section 7.10). Consider the regression

$$y = X\beta + D\eta + u \quad (5)$$

where D is a matrix of cluster dummy variables. If s^2 denotes the usual OLS estimate of the error variance and $\hat{\sigma}_\eta^2$ is the sample variance of the elements of $\hat{\eta}$, then a natural (but biased) estimate of ρ is $\hat{\rho} = \hat{\sigma}_\eta^2 / (s^2 + \hat{\sigma}_\eta^2)$. This estimator is based on the assumption that the error terms are equicorrelated within each cluster with correlation ρ .

An alternative way to make the degrees-of-freedom parameter a function of the data, originally proposed by Bell and McCaffrey (2002), was modified by Imbens and Kolesár (2016), and yet another degrees-of-freedom correction has very recently been proposed by Young (2016). Simulation results in Cameron and Miller (2015) suggest that the Imbens and Kolesár degrees-of-freedom parameter can often be very similar to the G^* parameter of CSS. Bell and McCaffrey (2002) also proposed a modified CRVE similar in spirit to the HC2 heteroskedasticity-consistent covariance matrix estimator of MacKinnon and White (1985). Unfortunately, calculating this CRVE involves finding the inverse symmetric square roots of $N_g \times N_g$ matrices for $g = 1, \dots, G$. Calculating the degrees-of-freedom parameter requires even more extensive computations. In view of the large sample sizes in most of our experiments (extremely large for the placebo laws experiments of Section 7 and the empirical example of Section 8), it was not feasible to study these procedures.² Another approach to inference in model (1) was suggested by Ibragimov and Müller (2010), but it requires that the coefficient of interest be separately identifiable from the data for each cluster, which is not the case in most of our experiments.

² For the state-size experiments of Section 4, in which $N = 2000$ and the largest N_g equals 242, simply calculating the Bell-McCaffrey CRVE would have taken about 17 times as much CPU time as bootstrapping with $B = 399$.

4. SIMULATION DESIGN: CONTINUOUS REGRESSORS

In this section, we use Monte Carlo simulation experiments to explore the implications of assumptions A1 and A3 for clustered data when the regressors are continuous. We study conventional inference based on the CRVE, wild bootstrap inference and inference where the critical values depend on G^* .

There are four sets of simulations: two with 50 clusters and two with 100 clusters. In each case, one set has equal-sized clusters, while the other has clusters with sizes proportional to the US states without the District of Columbia. Because the use of 50 clusters satisfies the ‘rule of 42’, we would expect to see reliable inference in all cases if the rule actually held.

The model we estimate is

$$y_{ig} = \beta_1 + \beta_2 X_{ig} + \epsilon_{ig}, \quad i = 1, \dots, N_g, \quad g = 1, \dots, G \quad (6)$$

where there are G clusters, and the g th cluster has N_g observations. Both the X_{ig} and the ϵ_{ig} are standard normal and uncorrelated across clusters. The within-cluster correlation is ρ_x for the X_{ig} and ρ_ϵ for the ϵ_{ig} . We do not allow ρ_ϵ to equal 1, but we do allow $\rho_x = 1$. In that case, the regressor is constant within each cluster—a situation that is commonly encountered in practice. We vary both ρ_x and ρ_ϵ across the simulations. Each simulated dataset has 2000 observations, and each experiment involves 400,000 replications.³ For the wild cluster bootstrap, we use 399 bootstrap samples.

Each simulation proceeds as follows:

1. Specify $\rho_x \in \{0, 0.2, \dots, 0.8, 1\}$ and $\rho_\epsilon \in \{0, 0.1, \dots, 0.8, 0.9\}$.
2. For each simulated sample, generate X_{ig} and ϵ_{ig} , use equation (6) to compute y_{ig} , with $\beta_2 = 0$, and then estimate equation (6) by OLS.
3. Test the hypothesis that $\beta_2 = 0$, using either t tests based on the CRVE with several different degrees-of-freedom parameters or a wild bootstrap test.
4. Repeat the two preceding steps 400,000 times, and estimate the rejection frequency of each test at the 0.05 level.

We compare the CRVE t statistic with six different critical values: $t(N - 2)$, $t(G - 1)$, $t(G^* - 1)$ for two values of G^* , and $t(G^*)$ for two values of G^* . The two values of G^* are based on $\rho = 0.99$ (because CSS suggest using $\rho = 1$, which sometimes caused numerical problems) and $\rho = \hat{\rho}$, which was defined just after equation (5). For reasons of space, however, we report results only for $t(G - 1)$ and $t(G^* - 1)$ based on $\hat{\rho}$. Using $t(N - 2)$ critical values always led to more severe over-rejection than using $t(G - 1)$ ones. In most cases, basing G^* on $\rho = \hat{\rho}$ worked better than basing it on $\rho = 0.99$.

Table I presents results for samples of 2000 observations spread equally across 50 clusters, so that assumption A3 holds.⁴ The reliability of CRVE inference with $t(G - 1)$ critical values depends on both ρ_ϵ and ρ_x . When ρ_x is close to 0, rejection rates are close to the desired 0.05 level. As ρ_x gets closer to 1, however, they increase, always exceeding 0.065 when $\rho_x = 1$. In general, increasing ρ_ϵ increases rejection rates slightly, although to a lesser degree than increasing ρ_x . The impact is most severe when ρ_x is large but less than 1.

Using critical values from the $t(G^* - 1)$ distribution frequently results in more accurate inferences, but there is a tendency to over-reject when ρ_x is large and to under-reject when ρ_ϵ is large. Using $t(G^*)$ instead of $t(G^* - 1)$ makes the over-rejection in the former case more severe and the under-rejection in the latter case less severe. When $\rho_x = 1$, G^* is invariant to the value of ρ . Except in this case,

³ Results for samples of 1000 observations (20 per cluster instead of 40), not reported, were almost identical to those for $G = 50$ reported here. These and other experiments suggest that, for all the simulation designs in this paper, rejection frequencies for given G are essentially invariant to N .

⁴ More complete versions of the tables in this section, which include results for all values of ρ_ϵ between 0.0 and 0.9 at intervals of 0.1, may be found in Section A.2 of the online Appendix.

Table I. Rejection frequencies with 50 equal-sized clusters

ρ_ϵ		ρ_x					
		0.0	0.2	0.4	0.6	0.8	1.0
0.0	$t(G-1)$	0.0503	0.0503	0.0508	0.0530	0.0572	0.0658
	$t(G^*-1)$	0.0500	0.0500	0.0496	0.0498	0.0501	0.0535
	Bootstrap	0.0501	0.0498	0.0497	0.0503	0.0505	0.0494
0.3	$t(G-1)$	0.0506	0.0505	0.0524	0.0565	0.0619	0.0665
	$t(G^*-1)$	0.0502	0.0495	0.0489	0.0498	0.0520	0.0543
	Bootstrap	0.0501	0.0495	0.0494	0.0502	0.0501	0.0502
0.6	$t(G-1)$	0.0511	0.0520	0.0541	0.0586	0.0630	0.0665
	$t(G^*-1)$	0.0495	0.0478	0.0462	0.0483	0.0516	0.0544
	Bootstrap	0.0502	0.0505	0.0504	0.0501	0.0497	0.0503
0.9	$t(G-1)$	0.0500	0.0508	0.0543	0.0592	0.0633	0.0655
	$t(G^*-1)$	0.0430	0.0416	0.0433	0.0480	0.0515	0.0537
	Bootstrap	0.0495	0.0497	0.0498	0.0505	0.0498	0.0496

Note: Rejection frequencies at the 0.05 level are based on 400,000 replications. There are 2000 observations. The effective number of clusters is $G^*(\hat{\rho})$. Wild bootstrap P -values are based on 399 bootstraps using the Rademacher distribution.

Table II. Rejection frequencies with 50 state-sized clusters

ρ_ϵ		ρ_x					
		0.0	0.2	0.4	0.6	0.8	1.0
0.0	$t(G-1)$	0.0594	0.0589	0.0597	0.0628	0.0673	0.0821
	$t(G^*-1)$	0.0514	0.0467	0.0368	0.0297	0.0259	0.0255
	Bootstrap	0.0504	0.0497	0.0497	0.0508	0.0508	0.0507
0.3	$t(G-1)$	0.0588	0.0636	0.0712	0.0810	0.0945	0.1053
	$t(G^*-1)$	0.0509	0.0447	0.0339	0.0298	0.0300	0.0297
	Bootstrap	0.0502	0.0510	0.0516	0.0515	0.0519	0.0516
0.6	$t(G-1)$	0.0585	0.0663	0.0755	0.0864	0.0987	0.1065
	$t(G^*-1)$	0.0484	0.0335	0.0251	0.0252	0.0281	0.0289
	Bootstrap	0.0502	0.0510	0.0521	0.0525	0.0522	0.0512
0.9	$t(G-1)$	0.0566	0.0673	0.0765	0.0884	0.0994	0.1073
	$t(G^*-1)$	0.0377	0.0233	0.0210	0.0237	0.0265	0.0288
	Bootstrap	0.0514	0.0520	0.0525	0.0528	0.0521	0.0510

Note: See note to Table I.

setting $\rho = 0.99$ results in (often substantially) lower values of G^* than using $\hat{\rho}$, which tends to cause noticeable under-rejection.

Even better results are obtained by using the wild bootstrap, which performs perfectly for all practical purposes. The smallest rejection frequency out of the 60 reported in the full version of Table I is 0.04942, and the largest is 0.05081. These numbers are close enough to 0.05 to be explained by chance. The standard error when the true rejection frequency is 0.05 is 0.000345, so the largest implied t -statistic for the hypothesis that the rejection frequency is 0.05 is less than 2.35.

Since assumption A3 holds, it is not surprising that the ‘rule of 42’ nearly holds in these simulations. Table II presents results from a second set of experiments in which that assumption is severely violated.

Cluster sizes are now roughly proportional to US state populations; the smallest clusters have just four observations, and the largest has 242. Even when ρ_ϵ and ρ_x are 0, the rejection rate is nearly 0.06. At the other extreme, when $\rho_\epsilon = 0.9$ and $\rho_x = 1$, the rejection rate is 0.1073. Increasing ρ_x leads to an increase in rejection rates. So does increasing ρ_ϵ , except when $\rho_x = 0$. Thus, with even modest amounts of intra-cluster correlation, the ‘rule of 42’ fails to hold in these experiments.

With state-sized clusters, using $t(G^* - 1)$ critical values generally results in under-rejection, which is quite severe when ρ_x is large and generally becomes worse as ρ_ϵ gets larger. The under-rejection is even more severe when G^* is based on $\rho = 0.99$ instead of $\rho = \hat{\rho}$. In the worst case ($\rho_\epsilon = 0$, $\rho_x = 0.6$), the rejection rate (not reported in the table) is just 0.0166. In this case, the average value of $G^*(\hat{\rho})$ is 9.43, while the average value of $G^*(0.99)$ is 5.97. Both these numbers seem to be unrealistically low. In cases such as these, using $t(G^*)$ instead of $t(G^* - 1)$ reduces the under-rejection only modestly.

As before, much better inferences are obtained by using the wild bootstrap, although it does not work quite as well as with equal-sized clusters. Rejection frequencies range from 0.0497 to 0.0528. There is thus a very modest tendency to over-reject in some cases.

In order to investigate assumption A1, we repeated both sets of experiments using 100 clusters instead of 50, holding the sample size constant at 2000. Table A.3 of the online Appendix reports results for 100 equal-sized clusters. The wild bootstrap works perfectly, except for simulation error. The other methods work better than they did with 50 clusters, but they tend to over-reject or under-reject in the same cases. Table A.4 reports results for 100 clusters that are roughly proportional to US state populations, with each state appearing twice. The two smallest clusters have just two observations, and the two largest have 121. In effect, there are two Californias, two Delawares, and so on. All methods work better than they did in Table II, but CRVE rejection frequencies can still be greater than 0.085. The wild cluster bootstrap rejection rate never exceeds 0.0514, although it does over-reject more often than it under-rejects.

These results demonstrate that inference based on the CRVE may not be reliable when cluster sizes differ substantially. Comparing G^* with G seems to provide valuable evidence that inference based on $t(G - 1)$ may be unreliable, but using critical values from the $t(G^* - 1)$ distribution does not always solve the problem. The most reliable approach in this setting, especially when G^* is small, is apparently to use the wild cluster bootstrap.

5. SIMULATION DESIGN: TREATMENT EFFECTS

Many applications to clustered data involve treatment effects, either at the cluster level or by time period within some clusters. We conducted three sets of experiments to investigate this type of application. Results for one set are reported here, and results for a second set are reported in Section A.3 of the online Appendix. The third set produced results that are not reported because they were visually indistinguishable from those of the first set.

In many empirical studies, only some observations in some clusters are treated. If i indexes individuals, g indexes jurisdictions, such as states, and t indexes time periods, then a classic ‘difference in differences’ (or ‘DiD’) regression can be written as

$$y_{igt} = \beta_1 + \beta_2 GT_{igt} + \beta_3 PT_{igt} + \beta_4 GT_{igt} PT_{igt} + \epsilon_{igt} \quad (7)$$

for $i = 1, \dots, N_g$, $g = 1, \dots, G$, and $t = 1, \dots, T$. Here GT_{igt} is a ‘cluster treated’ dummy that equals 1 if cluster g is ever treated, and PT_{igt} is a ‘period treated’ dummy that equals 1 if treatment occurs in time period t . The coefficient of most interest is β_4 , which shows the effect on treated clusters in periods when there is treatment. We let G_1 denote the number of treated clusters and M_1 denote the number of treated observations, i.e. observations for which $GT_{igt} PT_{igt} = 1$.

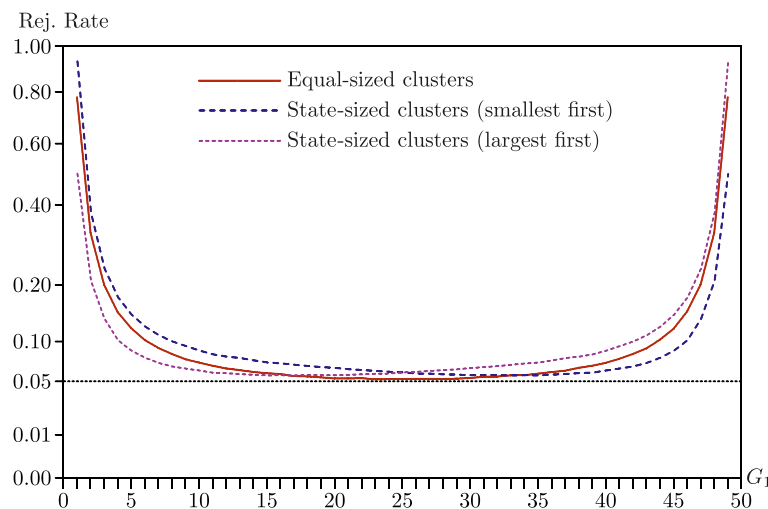


Figure 1. Rejection rates and proportion treated, DiD, $t(G - 1)$. [Colour figure can be viewed at wileyonlinelibrary.com]

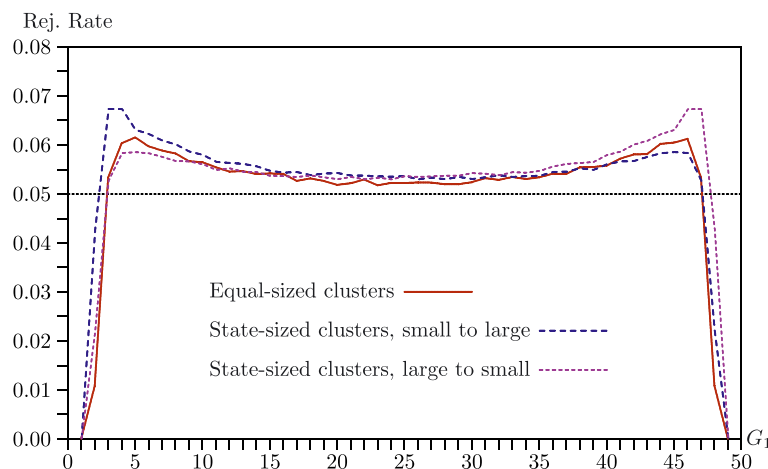


Figure 2. Rejection rates and proportion treated, DiD, $t(G^* - 1)$. [Colour figure can be viewed at wileyonlinelibrary.com]

In the DiD regression model, there is no role for ρ_x , and ρ_ϵ usually seems to have very little effect on rejection frequencies; we set the latter to 0.05 in all the experiments.⁵ As we show in Section 6, what matters is G_1 , the number of treated groups. In Figures 1–3, we report results for 50 clusters with 2000 observations. The treatments are applied to state-sized clusters both from smallest to largest and from largest to smallest. The simulations used 400,000 replications.

Figure 1 shows results for $t(G - 1)$ critical values. It is based on equation (7) with either zero or half of the observations in each cluster treated.⁶ Thus $PT_{igt} = 1$ for half the observations in each cluster, while $GT_{igt} = 1$ for G_1 clusters, with G_1 varying between 1 and 49. The vertical axis has been subjected to a square root transformation in order to accommodate both large and small rejection frequencies.

⁵ In early versions of the paper, we set $\rho_\epsilon = 0.5$, a number that is unrealistically high.

⁶ In exploratory experiments with fewer simulations, very similar results were obtained when either one quarter or three quarters of the observations in each cluster were treated.

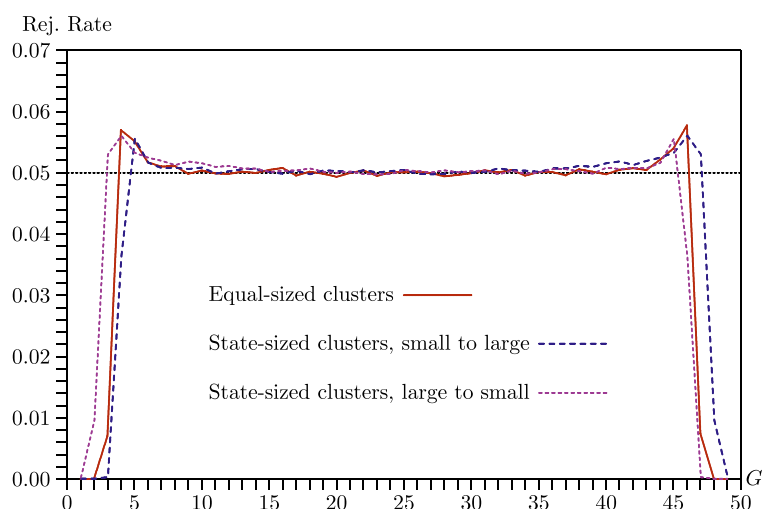


Figure 3. Rejection rates and proportion treated, DiD, wild bootstrap. [Colour figure can be viewed at wileyonlinelibrary.com]

There is very severe over-rejection in Figure 1 when either G_1 or $G - G_1$ is close to 0. This result is consistent with Monte Carlo results in Bell and McCaffrey (2002) and Conley and Taber (2011), and we explain the result in Section 6. The latter paper develops procedures for inference when there are just a few treated groups; see also MacKinnon and Webb (2016), which studies those procedures and develops additional ones. A very different procedure for inference when there is only one treated group has been proposed by Abadie *et al.* (2010) based on the idea of ‘synthetic controls’.

With equal-sized clusters, rejection frequencies are very close to 0.05 for G_1 between 17 and 34. With state-sized clusters, they tend to be somewhat higher. The graph for equal-sized clusters is symmetric around $G_1 = 25$, while those for state-sized clusters are quite asymmetric. For small values of G_1 , there is very severe over-rejection when the smallest clusters are treated first. For large values of G_1 , there is still serious over-rejection, but it is considerably less severe.

The results when the largest clusters are treated first are the mirror image of the results when the smallest clusters are treated first. This must be the case, because the absolute value of the t -statistic for $\beta_4 = 0$ in regression (7) is the same when the number of clusters treated is G_1 as it is when that number is $G - G_1$. We may conclude that over-rejection tends to be most severe when $\min(G_1, G - G_1)$ is small and the observations that are in the minority are from the smallest clusters. These results will be explained in Section 6.

Figure 2 shows results for $t(G^* - 1)$ critical values. In all cases, G^* is invariant to ρ . Except when the number of treated clusters is very small or very large, the tests always over-reject, but they never do so severely. There is extreme under-rejection when $G_1 = 1$ and $G_1 = 49$, because G^* is not much greater than 1 in those cases, and the Student's t -distribution has extremely long tails when the degrees of freedom parameter is very close to zero. Rejection frequencies are extremely sensitive to that parameter; using critical values based on $t(G^*)$ instead of $t(G^* - 1)$ can lead to severe over-rejection.

Figure 3 shows results for wild bootstrap tests based on simulations with 399 bootstraps. In all cases, there is severe under-rejection when either G_1 or $G - G_1$ is very small. For equal-sized clusters, there is modest over-rejection when G_1 and $G - G_1$ are both either 4 or 5, but the wild bootstrap tests work extremely well for G_1 between about 7 and 43. For state-sized clusters, the range of under-rejection is narrower (wider) for small G_1 , and wider (narrower) for large G_1 , when the largest (smallest) clusters are treated first. All of these results will be explained in Section 6.

It is common to allow for cluster fixed effects in DiD models by dropping the constant term and the GT_{ig} variable and adding G dummy variables, one for each cluster. This is necessary if treatment occurs at different times for different clusters. We performed a second set of experiments for the fixed-effects DiD model and obtained results that were visually indistinguishable from those in Figures 1–3, and are therefore not reported.

Another possibility is a pure treatment model, where there is no time dimension and the test regressor is an indicator variable that equals 1 for some number of clusters. Thus, for each cluster, either all observations are treated or all are not treated. We performed a third set of experiments for this model and obtained results that are similar to, but do differ in some respects from, those in Figures 1–3. They are reported in Section A.3 of the online Appendix. We also report some additional simulation results for the DiD case with between 12 and 32 clusters and varying degrees of inequality in cluster sizes in Section A.7. Even with very small numbers of clusters, the (restricted) wild cluster bootstrap seems to perform well provided that neither G_1 nor $G - G_1$ is too small.

6. WHY CRVE T -TESTS AND WILD BOOTSTRAP TESTS CAN FAIL

As we have seen, t -tests based on CRVE standard errors tend to over-reject, often very severely, when the number of treated clusters is very small or very large, and wild bootstrap tests based on restricted residuals tend to under-reject just as severely. In this section, we explain these phenomena. For simplicity, we focus on the pure treatment model:

$$y_{ig} = \beta_1 + \beta_2 d_{ig} + \epsilon_{ig} \quad (8)$$

where d_{ig} equals 1 for the first G_1 clusters and 0 for the remaining $G_0 = G - G_1$ clusters. Including additional regressors, or allowing only some observations within treated clusters to be treated, would not change the analysis in any fundamental way.

We make the simplifying assumption that G and N tend to infinity at the same rate, which implies that the N_g do not grow or shrink systematically as N increases. CRVE-based inference is known to be valid under weaker assumptions that allow G to grow more slowly than N , but not too slowly; see CSS. Since the focus of this section is on cases where CRVE-based inference fails despite a relatively strong assumption, there is no reason to relax that assumption. Moreover, many of our results do not require any asymptotic assumptions. They hold, at least qualitatively, in finite samples.

Standard asymptotic analysis is based on the assumption that the limit of $\phi \equiv G_1/G$ as $N \rightarrow \infty$ is strictly between 0 and 1. An alternative assumption, which is much more realistic when G_1 is small, is that G_1 is fixed as $N \rightarrow \infty$. This implies that $\phi \rightarrow 0$. This assumption allows us to explain all the simulation results in Figures 1 and 3, as well as ones reported later in this section, in Section 7, and in the online Appendix.

Equation (8) may be rewritten in vector notation as $\mathbf{y} = \beta_1 \mathbf{1} + \beta_2 \mathbf{d} + \boldsymbol{\epsilon}$, where $\mathbf{y}$, $\mathbf{1}$, $\mathbf{d}$ and $\boldsymbol{\epsilon}$ are N -vectors with typical elements y_{ig} , 1, d_{ig} and ϵ_{ig} , respectively, and i varies more rapidly than g . Then the OLS estimator of β_2 is

$$\hat{\beta}_2 = \frac{(\mathbf{d} - \bar{d}\mathbf{1})'\mathbf{y}}{(\mathbf{d} - \bar{d}\mathbf{1})'(\mathbf{d} - \bar{d}\mathbf{1})} = \beta_2 + \frac{(\mathbf{d} - \bar{d}\mathbf{1})'\boldsymbol{\epsilon}}{(\mathbf{d} - \bar{d}\mathbf{1})'(\mathbf{d} - \bar{d}\mathbf{1})} \quad (9)$$

where $\bar{d}$ denotes the sample mean of the d_{ig} . For convenience, let M_1 denote the number of treated observations and $M_0 = N - M_1$ denote the number of untreated observations, so that $\bar{d} = M_1/N$. For any finite N , $\bar{d}$ will generally differ from ϕ unless $N_g = N/G$ for all g , but, like ϕ , it must be $O(N^{-1})$ when G_1 is fixed.

The estimator $\hat{\beta}_2$ is inconsistent under the assumption that G_1 is fixed. Since the denominator of $\hat{\beta}_2$ is $\bar{d}(1 - \bar{d})N = M_1 M_0 / N$, we can use equation (9) to obtain

$$\hat{\beta}_2 - \beta_2 = \frac{1}{M_1} \sum_{g=1}^{G_1} \sum_{i=1}^{N_g} \epsilon_{ig} - \frac{1}{M_0} \sum_{g=G_1+1}^G \sum_{i=1}^{N_g} \epsilon_{ig} \quad (10)$$

The first term in equation (10) is $1/M_1$ times a summation of M_1 mean-zero random variables. Thus it is $O_p(M_1^{-1/2}) = O_p(1)$. The second term is $1/M_0$ times a summation of $M_0 = N - M_1$ mean-zero random variables. Thus it is $O_p(N^{-1/2})$. We conclude that the first term, even though it depends on just $M_1 = O(1)$ observations, must be larger asymptotically than the second term. Thus a fundamental condition for consistency, namely that the impact of any one observation on $\hat{\beta}_2$ must tend to zero as $N \rightarrow \infty$, is violated.

From expression (2), it is evident that the CRVE for $\hat{\beta}_2$ is proportional to

$$\frac{\sum_{g=1}^G (d_g - \bar{d} \iota_g)' \hat{\epsilon}_g \hat{\epsilon}_g' (d_g - \bar{d} \iota_g)}{((d - \bar{d} \iota)'(d - \bar{d} \iota))^2} \quad (11)$$

The CRVE would provide a good estimate of $\text{var}(\hat{\beta}_2)$ if its numerator provided a good estimate of $(d - \bar{d} \iota)' \Omega (d - \bar{d} \iota)$. However, it is impossible to estimate $\text{var}(\hat{\beta}_2)$ consistently when G_1 is fixed; this is a special case of Corollary 1 in CSS.

Equations (9) and (11) imply that, under the null hypothesis, the t -statistic for $\beta_2 = 0$ is

$$t_2 = \frac{c(d - \bar{d} \iota)' \epsilon}{\left(\sum_{g=1}^G (d_g - \bar{d} \iota_g)' \hat{\epsilon}_g \hat{\epsilon}_g' (d_g - \bar{d} \iota_g) \right)^{1/2}} \quad (12)$$

where c is the square root of $((G - 1)(N - 2)) / (G(N - 1))$. This test statistic has variance very much greater than that of the $t(G - 1)$ distribution when G_1 (or G_0) is small. The reason is that its denominator, which is the square root of the numerator of expression (11), is not only inconsistent but also severely biased downwards in that case.

In scalar notation, $1/c$ times the numerator of expression (11) can be written as

$$(1 - \bar{d})^2 \sum_{g=1}^{G_1} \left(\sum_{i=1}^{N_g} \hat{\epsilon}_{ig} \right)^2 + \bar{d}^2 \sum_{g=G_1+1}^G \left(\sum_{i=1}^{N_g} \hat{\epsilon}_{ig} \right)^2 \quad (13)$$

This expression is supposed to estimate the quantity

$$\begin{aligned} (d - \bar{d} \iota)' \Omega (d - \bar{d} \iota) &= (1 - \bar{d})^2 \sum_{g=1}^{G_1} \iota' \Omega_g \iota + \bar{d}^2 \sum_{g=G_1+1}^G \iota' \Omega_g \iota \\ &= (1 - \bar{d})^2 \sum_{g=1}^{G_1} \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} \omega_{ij}^g + \bar{d}^2 \sum_{g=G_1+1}^G \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} \omega_{ij}^g \end{aligned} \quad (14)$$

where ω_{ij}^g denotes the ij th element of the covariance matrix $\mathbf{\Omega}_g$. When $G_1 = 1$, the residuals for cluster 1 must sum to zero, because $d_{i1} = 1$ and $d_{ig} = 0$ for $g > 1$. This implies that the first term in expression (13) equals zero. In contrast, the first term in expression (14) is not zero. It will generally be quite large relative to the second term, because $\bar{d}^2$ must be very small when $M_1 \ll M_0$. We conclude that equation (13) will always underestimate equation (14) when $G_1 = 1$. For large G , the underestimation must be severe, because the first term in equation (14) is $O(1)$ and the second is $O(N^{-1})$. Because $\bar{d}$ depends on the size of the treated clusters, the over-rejection will tend to be more severe when the smallest clusters are treated than when the largest clusters are treated, which is exactly what we see in Figure 1.

When two or more clusters are treated, the residuals for each treated cluster will not sum to zero, but they must sum to zero over all the treated clusters. Thus the sum of squared summations in the first term of equation (13) will always underestimate the corresponding double summation in equation (14). In Section A.4 of the online Appendix, we show that when the errors are i.i.d. the expectation of the squared summation for the first treated cluster underestimates the corresponding true variance by a factor of $(M_1 - N_1)/M_1$. Thus the underestimation should go away as G_1 increases. In typical cases where N_1/M_1 tends to zero fairly rapidly, it does indeed go away quite quickly, as Figure 1 illustrates.

As Figure 3 shows, the restricted wild cluster bootstrap does not solve this problem. For the dummy variable regression (8), the restricted wild cluster bootstrap DGP (3) is

$$y_{ig}^{*j} = \tilde{\beta}_1 + \tilde{\epsilon}_{ig} v_g^{*j} \quad (15)$$

where $\tilde{\beta}_1$ is the sample mean of the y_{ig} , and $\tilde{\epsilon}_{ig} = y_{ig} - \tilde{\beta}_1$. The OLS estimate $\hat{\beta}_2^{*j}$ for the j th bootstrap sample is then given by equation (9), with ϵ replaced by ϵ^{*j} , a vector with typical element $\tilde{\epsilon}_{ig} v_g^{*j}$, and the bootstrap t -statistic is

$$t_2^{*j} = \frac{c(\mathbf{d} - \bar{d}\mathbf{1})'\epsilon^{*j}}{\left(\sum_{g=1}^G (\mathbf{d}_g - \bar{d}\mathbf{1}_g)' \hat{\epsilon}_g^{*j} (\hat{\epsilon}_g^{*j})' (\mathbf{d}_g - \bar{d}\mathbf{1}_g)\right)^{1/2}} \quad (16)$$

where $\hat{\epsilon}_g^{*j}$ is the subvector of OLS residuals for cluster g and bootstrap sample j , and c is the square root of $((G-1)(N-2))/(G(N-1))$.

Now consider the extreme case in which $G_1 = 1$. The numerator of equation (16) becomes

$$c(1 - \bar{d}) \sum_{i=1}^{N_1} \epsilon_{i1}^{*j} - c\bar{d} \sum_{g=2}^G \sum_{i=1}^{N_g} \epsilon_{ig}^{*j} \quad (17)$$

Because $\bar{d} = N_1/N$, the first term in expression (17) must be the dominant one asymptotically, and also in finite samples unless N_1 is extraordinarily large. For the Rademacher distribution, the vectors of bootstrap error terms for $g = 1$ can have just two values, namely $\tilde{\epsilon}_1$ and $-\tilde{\epsilon}_1$. Therefore, when ϵ_1 is such that $|t_2|$ is large, $\tilde{\epsilon}_1^{*j}$ will probably be such that $|t_2^{*j}|$ is large. The distribution of the bootstrap t statistics t_2^{*j} is then bimodal, with half the realizations in the neighborhood of t_2 and the other half in the neighborhood of $-t_2$. Thus when $|t_2|$ is large, $|t_2^{*j}|$ tends to be even larger for a substantial number of bootstrap samples, making it rare to obtain a bootstrap P -value below any specified level for a test. This explains why the wild cluster bootstrap under-rejects severely when $G_1 = 1$.⁷

⁷ This argument implicitly assumes that the distributions of t_2 and the t_2^{*j} are similar, which may not be true if the disturbances for the treated and untreated observations have very different variances. In such cases, the wild cluster bootstrap would almost certainly not work well, but it might not always under-reject.

Figure 4 illustrates this phenomenon. The solid line is a kernel estimate based on 5 million replications of the empirical density of the statistic t_4 for the DiD model of equation (7) when $G_1 = 1$; recall that t_4 in the DiD model (7) plays the same role as t_2 in the pure treatment model (8). There are 2000 observations and 50 equal-sized clusters. This density is extremely dispersed. The 0.975 quantile is 14.226, whereas that for the $t(49)$ distribution is 2.010. Not surprisingly, the CRVE $t(G - 1)$ test rejects 77.75% of the time at the .05 level.

The dotted line in Figure 4 shows kernel density estimates of the bootstrap t statistics for a particular sample in which the realized value $\hat{t}_4 = 14.226$, the 0.975 quantile. This distribution is extremely bimodal, and each half of it is quite asymmetrical. The dashed line is similar, but it shows the bootstrap distribution for a particular sample in which $\hat{t}_4 = 9.188$, the 0.90 quantile. The two halves of the bootstrap distribution for $\hat{t}_4 = 14.226$ are much further apart than the two halves of the bootstrap distribution for $\hat{t}_4 = 9.188$. The 0.975 quantiles are 16.01 in the former case and 10.94 in the latter.

Although the Rademacher distribution is responsible for the bimodality of the bootstrap distributions in Figure 4, using another auxiliary distribution would not solve the problem. Figure 5 shows what happens when the standard normal distribution is used as the auxiliary distribution. The density of t_4 is the same as in Figure 4. The two bootstrap densities are, once again, conditional on $\hat{t}_4 = 14.226$ and $\hat{t}_4 = 9.188$. Although these densities are not bimodal, they are much too dispersed and consequently provide very poor approximations to the density of t_4 . Moreover, they become more dispersed as $|t_4|$ increases.

The cases shown in Figure 4 are typical. As the test statistic becomes larger in absolute value, the bootstrap distributions tend to become more spread out. Thus it is not surprising that restricted wild bootstrap tests never reject at any conventional level of significance when $G_1 = 1$ in these experiments. No matter how large the absolute value of the test statistic, many of the realizations of the bootstrap statistic are even larger.

When $G_1 = 2$, there are four possible pairs of vectors of bootstrap error terms for the first two clusters. Two of these pairs are $[\tilde{\epsilon}_1, \tilde{\epsilon}_2]$ and $[-\tilde{\epsilon}_1, -\tilde{\epsilon}_2]$, which are very likely to yield large values of $|t_2^{*j}|$, while the other two are $[\tilde{\epsilon}_1, -\tilde{\epsilon}_2]$ and $[-\tilde{\epsilon}_1, \tilde{\epsilon}_2]$, which are not assured to do so. Thus the distribution of the t_2^{*j} for $G_1 = 2$ will still tend to have a large variance when $|t_2|$ is large, but not as large as when $G_1 = 1$. For the Rademacher distribution, the number of possible sets of bootstrap error terms for the treated observations is 2^{G_1} . As G_1 increases, the proportion of bootstrap samples

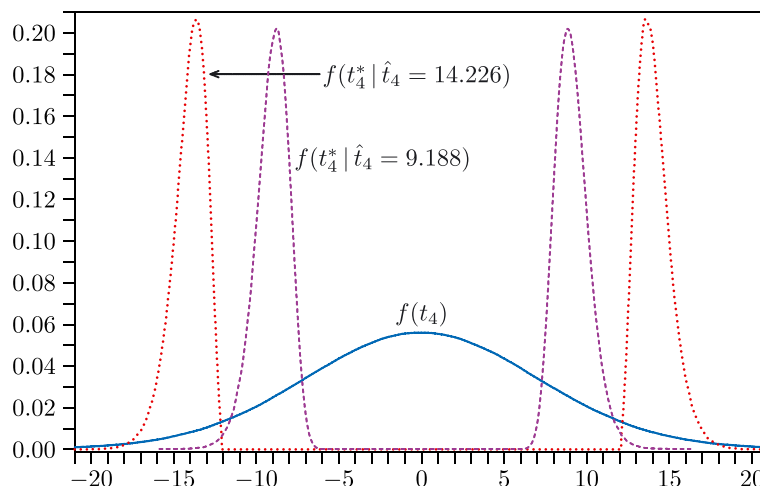


Figure 4. Densities of actual and bootstrap t -statistics when $G_1 = 1$. [Colour figure can be viewed at wileyonlinelibrary.com]

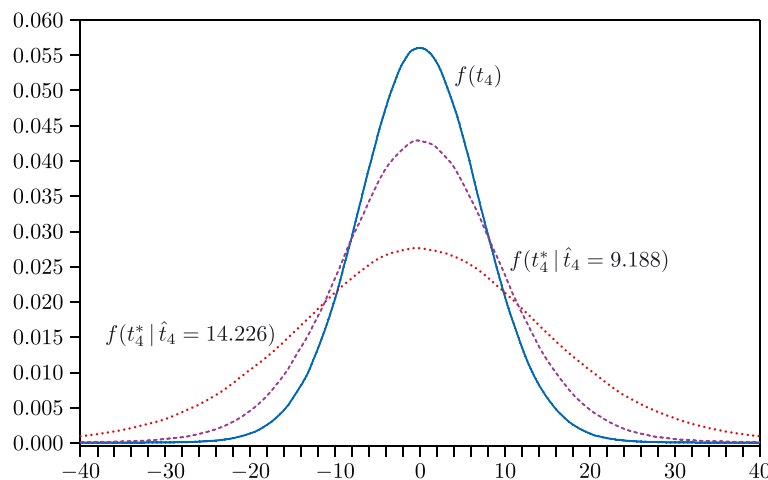


Figure 5. Densities of t -statistics for $G_1 = 1$ when auxiliary distribution is $N(0, 1)$. [Colour figure can be viewed at wileyonlinelibrary.com]

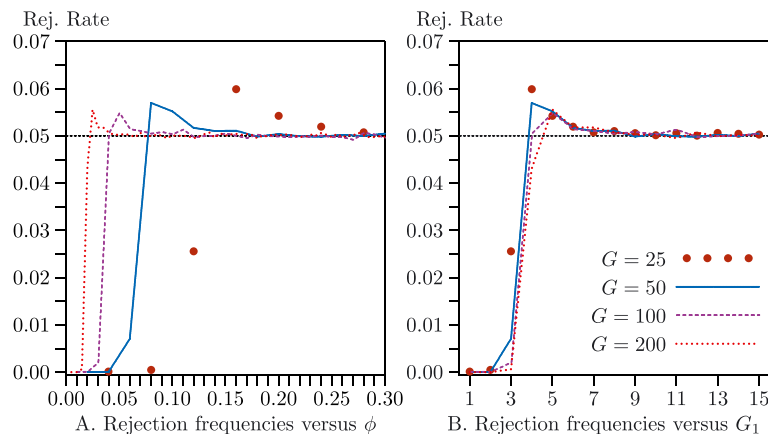


Figure 6. Restricted wild cluster bootstrap rejection. [Colour figure can be viewed at wileyonlinelibrary.com]

for which $\tilde{\epsilon}_g^{*j} = \hat{\epsilon}_g$ or $\tilde{\epsilon}_g^{*j} = -\hat{\epsilon}_g$ for all $g \leq G_1$ rapidly declines. Once G_1 becomes large enough, the problem goes away; see Sections A.5 and A.7 of the online Appendix.

The foregoing analysis implies that the key parameter is G_1 rather than ϕ . The next two figures confirm this. Figure 6 shows what happens for the case of equal-sized clusters in a DiD model. Here G takes the values 25, 50, 100 and 200, with $N_g = 40$ in the first three cases and (to save computer time) $N_g = 20$ in the last one. The left-hand panel shows rejection frequencies as a function of ϕ . It demonstrates that the range of values of ϕ for which the wild bootstrap yields accurate inferences becomes wider as G increases. The right-hand panel shows rejection frequencies as a function of G_1 . It suggests that the relationship between rejection frequencies and G_1 is almost invariant to G .

As we have seen, the numerator of the bootstrap test statistic under the null, $(\mathbf{d} - \bar{\mathbf{d}}\mathbf{t})' \epsilon^{*j}$, depends heavily on the restricted residuals for the treated clusters when G_1 is small. Thus it might seem that the failure of the wild cluster bootstrap for small G_1 could be avoided by using the unrestricted residuals $y_{ig} - X_{ig}\hat{\beta}$ rather than the restricted residuals $y_{ig} - X_{ig}\tilde{\beta}$ in the bootstrap DGP (3). Unfortunately, this is not so. Figure 7 deals with precisely the same cases as Figure 6, but the wild cluster bootstrap now uses unrestricted residuals. Instead of severe under-rejection for small values of G_1 , there is now very severe over-rejection.

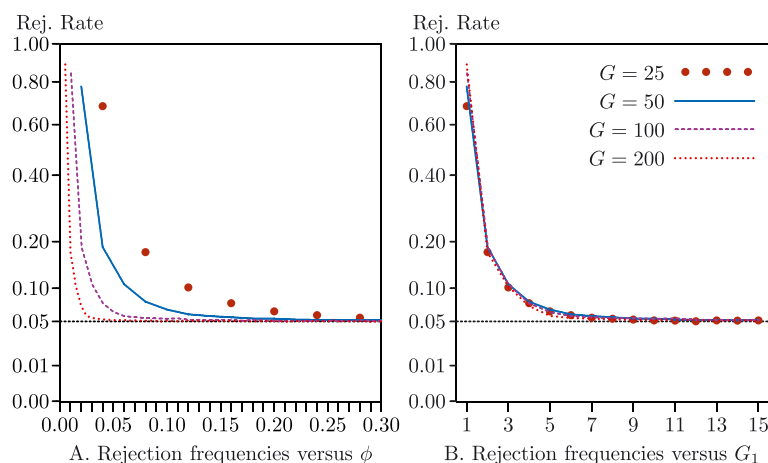


Figure 7. Unrestricted wild cluster bootstrap rejection frequencies. [Colour figure can be viewed at wileyonlinelibrary.com]

It is easy to explain the results in Figure 7. As before, the bootstrap test statistics t_2^{*j} depend mainly on the bootstrap error terms for the G_1 treated clusters. But the OLS residuals used to compute them are orthogonal to the treatment dummy. Therefore, when $G_1 = 1$, the average of the bootstrap error terms over the observations in the treated cluster must be zero. Thus the first term in expression (17) is zero, and the distribution of the t_2^{*j} depends almost entirely on the second term. Since the value of t_2 depends principally on the first term, it is not surprising that the bootstrap distribution provides a terrible approximation when $G_1 = 1$. Things improve as G_1 increases, but they actually get worse as G increases, because that makes the first term larger relative to the second one. Observe that, in Figure 7, the over-rejection for $G_1 = 1$ becomes more severe as G increases.

We conclude that, when G_1 is very small, using either CRVE t statistics or unrestricted wild cluster bootstrap tests is likely to result in severe over-rejection, while using restricted wild cluster bootstrap tests is likely to result in severe under-rejection. For given G_1 , all three tests tend to perform worse as $\bar{d}$, the fraction of treated observations, declines. Since adding additional untreated clusters actually reduces $\bar{d}$, all three tests must fail asymptotically when G_1 is held fixed as the sample size increases. For the pure treatment model analyzed in this section, and also for the DiD model (7), all the results discussed in this paragraph also hold if G_1 is replaced by G_0 and $\bar{d}$ is replaced by $1 - \bar{d}$. However, this is not true for DiD models in which treatment occurs at different times in different clusters; see Section 7.

The results in Figures 6 and 7 suggest that, for the DiD case with equal-sized clusters, the wild cluster bootstrap (either restricted or unrestricted) can probably be used safely when $7 \leq G_1 \leq G - 7$, but it is likely to be seriously misleading for values less than 4 or greater than $G - 4$. Of course, G_1 would probably need to be larger for the bootstrap to yield reliable results if the sizes of the treated clusters varied substantially; see Section A.7 of the online Appendix. It might well also need to be larger if there were pronounced heteroskedasticity that affected treated and untreated clusters differently, or if the model and dataset differed in other important ways from those in our experiments. Thus the exact value of G_1 needed for reliable inference is certain to be data-dependent.

7. PLACEBO LAWS

An alternative way to study the reliability of inference using clustered data is to use real-world data and simulate the effect of ‘placebo laws’. This ingenious approach was developed in Bertrand *et al.* (2004), hereafter referred to as BDM, which uses data from the US Current Population Survey. The

dependent variable is the log of weekly earnings for women aged 25–50 from 1979 to 1999. The authors note that there is an issue with the modest number of clusters, but they do not mention the potential issues with clusters of varying sizes. In fact, they report only the mean cluster size.

The regression for y , the natural logarithm of women's earnings, is

$$y = \beta_{\text{treat}} \text{TREAT} + \text{YEARS } \beta_{\text{years}} + \text{STATES } \beta_{\text{states}} + \text{OTHERS } \beta_{\text{other}} + \epsilon \quad (18)$$

where YEARS and STATES are full sets of fixed effects (so that there is no constant term), and OTHERS are age, age squared and a set of education dummy variables. The treatment variable is analogous to the interaction term in a standard DiD equation, where it would be set to 1 for observations in the treatment states during the treatment periods and to 0 otherwise. In regression (18), the treatment variable is instead set to 1 randomly, so that β_{treat} should be insignificantly different from zero. If the tests were working properly, we would expect β_{treat} to be significantly different from zero 5% of the time when testing at the 0.05 level.

The experiment in BDM is designed so that the treatment variable is randomly assigned to different states in each replication. For each replication, half the states are chosen at random to be treatment states and half to be controls. Also, for each replication, a year between 1985 and 1995 is chosen at random to be the beginning of the treatment period. If this year is called year^* , then the treatment variable is

$$\text{TREAT} = I(\text{state} = \text{treated}) I(\text{year} \geq \text{year}^*)$$

where $I(\cdot)$ is the indicator function. Since these treatment variables are assigned at random, they should on average have no estimated impact on earnings.

Our simulations are similar to, but more extensive than, those in BDM. Instead of treating half the states, we perform 51 sets of simulations, according to the number of states treated. There are 51 states because we include the District of Columbia. We omit observations for which earnings < \$20, which may be erroneous and are likely to have large residuals because of the log transformation, leaving us with a sample of 547,518 observations. In our principal experiments, we use micro data throughout, whereas most of the BDM simulations use data that are aggregated at the state–year level. In addition, the year that treatment starts is chosen at random for each treated state, rather than being the same for all treated states.

We estimate equation (18) for each replication and compute five different tests of the hypothesis that $\beta_{\text{treat}} = 0$. The first employs a t -statistic based on a heteroskedasticity robust standard error and the $N(0, 1)$ distribution. The second employs a t -statistic based on a cluster robust standard error and the $t(G - 1)$ distribution, as in the previous two sections. The third uses the same t -statistic with critical values from the $t(G^* - 1)$ distribution, with G^* based on an estimate of ρ . The fourth and fifth use P -values computed by the restricted and unrestricted wild cluster bootstrap techniques.

Results for the first two tests are based on 100,000 replications, where the states and years to be treated are chosen randomly with replacement. However, those for $t(G^* - 1)$ and the wild bootstrap tests are based on only 10,000 replications because of computational cost. When $G_1 = 1$, there are only $51 \times 11 = 561$ possible choices of state and years to treat. Therefore, results for that case are based on enumeration of those 561 cases.

The largest cluster (California) contains 42,625 observations. This makes it extremely expensive to compute G^* . A single replication using optimized code takes about 40 minutes. In order to make the experiments feasible, we compute G^* using only 1/100 of each sample. This reduces computational time by a factor of several thousand. Limited supporting experiments suggests that alternative

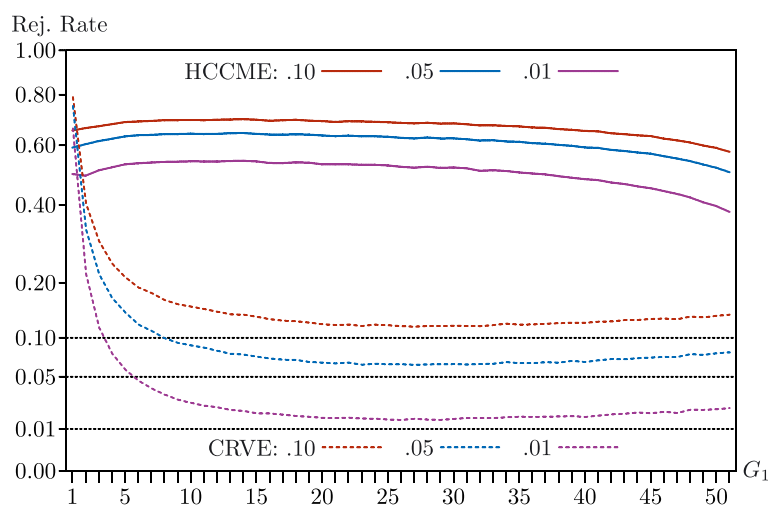


Figure 8. HCCME and CRVE rejection frequencies for placebo laws. [Colour figure can be viewed at wileyonlinelibrary.com]

approaches, such as using 1/50 samples or using several 1/100 samples and averaging, would have yielded almost identical results. The entire sample is used to estimate $\hat{\rho}$, which is always about 0.031 or 0.032.

Figure 8 plots rejection frequencies against G_1 , the number of states treated, for t tests based on both HCCME and CRVE standard errors. Even though intra-state correlations seem to be very low, using t -statistics based on HCCME standard errors results in very severe over-rejection. Rejection frequencies at the 0.05 level always exceed 0.50. These results imply that the error terms are not simply equicorrelated within clusters, as in Moulton (1990). If they were, the state fixed-effect dummies would soak up any state-specific random effects, and the remaining errors would be uncorrelated.

Using cluster robust standard errors and the $t(G - 1)$ distribution works much better than using heteroskedasticity robust standard errors, except when only one state is treated. Nevertheless, as might have been expected from the results in Sections 5 and 6, there is severe over-rejection when the number of treated states is small. For larger numbers of treated states, there is always over-rejection, but it is fairly modest. Rejection frequencies at the 0.05 level always exceed 0.063, and they exceed 0.07 about half the time. In contrast to the results in Figure 1, the results for many treated states are very different from the results for few treated states. The two cases are no longer equivalent because the years in which treatment starts vary randomly across treated states.

Figure 9 plots rejection frequencies against G_1 for restricted wild cluster bootstrap tests and for $t(G^* - 1)$ critical values. Both tests under-reject severely when G_1 is very small. For larger values of G_1 , the wild bootstrap performs extremely well, but using $t(G^* - 1)$ critical values leads to moderate over-rejection which increases with G_1 . In Section A.4 of the online Appendix, we report results for unrestricted wild bootstrap tests. They always reject more often than the restricted ones, and they over-reject severely when G_1 is small.

An alternative to using micro data is to average the values of all variables over all observations in each state–year pair, as BDM did in most of their experiments.⁸ This results in $51 * 21 = 1071$ observations and ensures that cluster sizes are perfectly balanced, with 21 observations in each. It also greatly reduces computational costs. Because the treatment variable in our experiments does not vary within state–year pairs, we might not expect much loss of power from this sort of aggregation. Evidence on this point is provided in Section A.9 of the online Appendix.

⁸ BDM took the average values of residuals, while we take average values of the actual data.

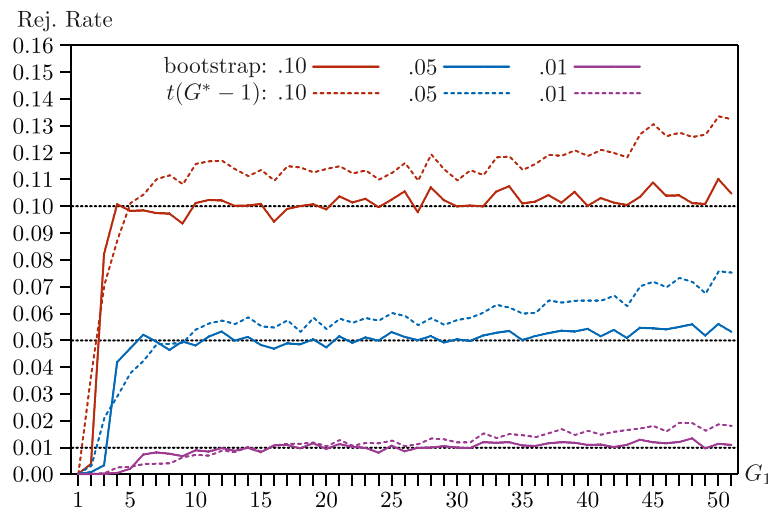


Figure 9. Bootstrap and $t(G^* - 1)$ rejection frequencies for placebo laws. [Colour figure can be viewed at wileyonlinelibrary.com]

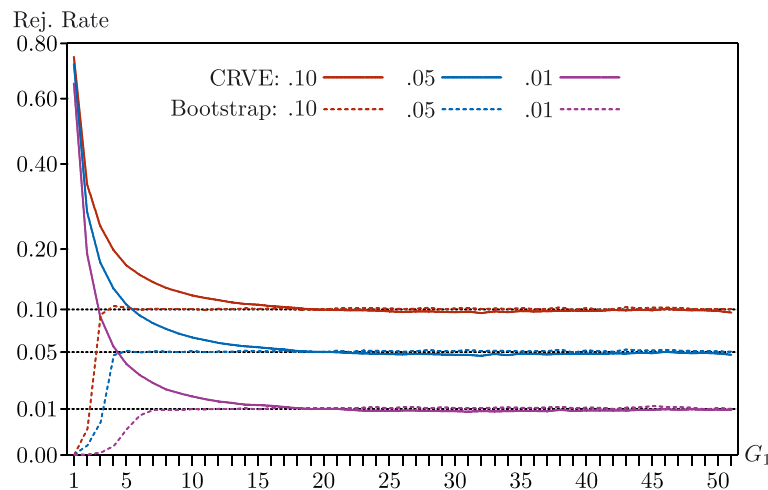


Figure 10. Rejection frequencies for placebo laws using aggregate data. [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 10 shows two sets of rejection frequencies as functions of G_1 for the aggregate placebo laws experiments. Additional results may be found in Section A.8. Results for cluster robust t -statistics using the $t(G - 1)$ distribution are based on 400,000 replications, and results for the wild cluster bootstrap are based on 100,000 replications. The CRVE t -statistics over-reject severely when G_1 is very small, but they perform very well for most values of G_1 and actually under-reject slightly in most cases. As usual, the (restricted) wild cluster bootstrap under-rejects severely when G_1 is very small, but it performs extremely well for $G_1 \geq 5$. The fact that both procedures perform better with aggregate data than with micro data reflects the fact that cluster sizes are always balanced in the former case.

8. EMPIRICAL EXAMPLE

The results in Sections 4, 5 and 7 have shown that inference based on the standard CRVE coupled with the $t(G - 1)$ distribution may be unreliable in situations with wildly different cluster sizes, but that other methods are more reliable. In this section, we illustrate how using either the wild cluster

bootstrap or the effective number of clusters can affect inference in practice by replicating a few select results from Angrist and Kugler (2008).⁹

Angrist and Kugler (2008) investigates the impact of an exogenous shock in the price of coca within Columbia on economic and criminal activity. To estimate economic outcomes, the paper uses data on individuals who live in various Columbian departments. We replicate select results from the paper's Table 6, which uses a DiD methodology to estimate the impact of increased coca prices on log hours worked. In this specification, the rural departments are 'treated' and the urban departments are 'controls', because coca production was concentrated in the rural areas. The equation can be written as

$$y_{ijt} = X_i' \mu + \beta_j + \delta_t + \alpha_{0,95-97} g_{jt,95-97} + \alpha_{0,98-00} g_{jt,98-00} + \alpha_{1,95-97} d_{jt,95-97} + \alpha_{1,98-00} d_{jt,98-00} + \epsilon_{ijt} \quad (19)$$

Here X_i is a vector of control variables, β_j is a department dummy, δ_t is a year dummy, the α_0 terms are DiD coefficients for the rural growing areas and the α_1 terms are DiD coefficients for the urban areas.

We replicate two sets of estimates from the paper's Table 6, namely, columns 6 and 9. The former explains log hours for men, and it excludes departments that are medium producers of cocoa from the analysis. The latter explains log hours for teenage boys, and it includes the medium producers. These regressions were chosen in part because the estimated P -values implicitly reported for the test regressors are quite low. The sample for men has 181,882 observations, and the sample for teenage boys has 22,141 observations. In both cases, the cluster sizes are wildly different. For the adult men, there are 32 clusters; the largest has 25,775 observations and the smallest has only 509. For the teenage boys, there are 38 clusters; the largest has 1920 observations and the smallest has only 42.

The results from these regressions can be found in Table III. Panel A replicates the results for men (column 6 of the original table). We report six P -values for each coefficient. The conventional ones use the $t(G-1)$ distribution. The symmetric bootstrap P -values are based on 99,999 bootstrap samples (far more than would normally be needed). There are two G^* P -values, one based on $t(G^*)$ and one based on $t(G^*-1)$, both estimated with $\hat{\rho} = 0.025$. All of the unconventional P -values are larger than the conventional ones, and many of them are much larger. Only the rural coefficient for 1998–2000, $\alpha_{1,98-00}$ in equation (19), is significant at the 5% level according to the bootstrap.

Panel B replicates the results for teenage boys (column 9 of the original table). In this case, the bootstrap P -values are very similar to the conventional ones, with both rural coefficients being significant at the 5% level. However, all of the $t(G^*-1)$ P -values, and three of the four $t(G^*)$ P -values, suggest that they are not significant. At 0.155, the estimate of ρ is much larger for the boys than for the men.

We also report estimates based on 288 aggregate observations at the department–year level for the men and 340 aggregate observations for the teenage boys. The estimates for the men, in Panel A, are of similar magnitude and sign to the micro estimates; however, all of the standard errors are significantly larger than before. The estimates for the boys, in Panel B, are in some cases quite different from the micro estimates. Two of the coefficients differ by a factor of almost three, and the standard errors are substantially larger.

Panel C calculates joint tests for the statistical significance of the two urban and the two rural coefficients. The reported statistics are quadratic forms in the 2-vectors of parameter estimates and the inverse of the appropriate 2×2 block of the CRVE. These Wald statistics are then divided by 2 so as to use the $F(2, G-1)$ distribution, as suggested in Cameron and Miller (2015), which also suggests computing wild cluster bootstrap P -values. One of the statistics, for rural men, appears to be highly significant based on its $F(2, G-1)$ P -value, but it is not significant at the 5% level according to the bootstrap. Because there is currently no way to calculate G^* for a joint test, we are unable to report results based on the effective number of clusters.

⁹ We thank Josh Angrist for making his data publicly available in his own data archive.

Table III. Empirical Example based on Angrist and Kugler (2008)

Panel A: log hours—adult men—no medium producers (32 departments)				
	Rural 95–97	Rural 98–00	Urban 95–97	Urban 98–00
Coeff.	0.0581	0.1219	0.0405	0.0740
SE	0.0278	0.0359	0.0193	0.0395
<i>t</i> -stat.	2.091	3.395	2.099	1.872
$G_{0.025}^*$	6.393	3.402	4.482	1.529
<i>P</i> -values				
$t(G - 1)$	0.045	0.002	0.044	0.071
$t(G_{0.025}^* - 1)$	0.079	0.035	0.096	0.240
$t(G_{0.025}^* - 1)$	0.087	0.059	0.114	0.447
Bootstrap	0.186	0.028	0.092	0.090
<i>Aggregate</i>				
Coeff.	0.0357	0.0975	0.0116	0.0563
SE	0.0621	0.0850	0.0455	0.0793
<i>t</i> -stat.	0.575	1.147	0.255	0.710
Panel B: log Hours—teenage Boys—all producers (38 departments)				
	Rural 95–97	Rural 98–00	Urban 95–97	Urban 98–00
Coeff.	0.1185	0.2150	−0.0040	−0.0472
SE	0.0519	0.1052	0.0680	0.0904
<i>t</i> -stat.	2.285	2.044	−0.058	−0.522
$G_{0.155}^*$	9.277	11.537	8.859	12.892
<i>P</i> -values				
$t(G - 1)$	0.028	0.048	0.954	0.605
$t(G_{0.155}^* - 1)$	0.047	0.064	0.955	0.611
$t(G_{0.155}^* - 1)$	0.051	0.067	0.955	0.611
bootstrap	0.030	0.050	0.958	0.628
<i>Aggregate</i>				
Coeff.	0.0444	0.1432	−0.0490	−0.1139
SE	0.0664	0.1314	0.0899	0.1243
<i>t</i> -stat.	0.669	1.090	−0.545	−0.916
Panel C: joint tests				
	Rural men	Rural boys	Urban men	Urban boys
Test stat.	5.830	2.614	2.402	0.279
$F(2, G - 1)$	0.007	0.087	0.107	0.758
bootstrap	0.091	0.140	0.226	0.796

9. CONCLUSION

This paper identifies two circumstances in which inferences based on cluster robust standard errors should not be trusted, even when the sample size is large and the ‘rule of 42’ is satisfied. There can be serious over-rejection when clusters are of wildly unequal sizes. With dummy regressors for treatment, there can be extremely severe over-rejection when the number of treated (or untreated) clusters is small, whether or not clusters are of equal sizes. This is true both for cluster-level treatments and for difference-in-differences regressions with and without cluster fixed effects. The problem is most severe when some or all of the treated clusters are small.

These results contrast with some of the earlier results of Bertrand *et al.* (2004) and Cameron *et al.* (2008), which suggest that cluster robust inference is reliable with 50 clusters. Those results are

misleading because they are based on equal-sized clusters and, in the former case, on aggregate data. Using aggregate data automatically solves the problem of unbalanced cluster sizes, but, as our empirical example illustrates, it can result in much larger standard errors.

Using critical values based on the effective number of clusters, as defined by Carter *et al.* (2017), instead of the actual number often improves matters substantially, although it does not always work as well as might be hoped. In contrast, with one notable exception, the restricted wild cluster bootstrap generally yields very reliable inferences for the cases we study. The exception is that, for reasons explained in Section 6, the restricted wild cluster bootstrap can under-reject very severely when only a few clusters are treated or untreated. All the methods that we study perform very badly in this case.

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